

Materials made of molecules

At the end of this chapter, you will have an understanding of the bonding in substances formed by the joining of two or more non-metal atoms. When non-metals bond with other non-metals, their atoms share electrons in order to gain stable valence (outer) shells.

Most substances formed from non-metallic elements have relatively low melting and boiling points and are composed of small molecules.

Science understanding

- the ability of atoms to form chemical bonds can be explained by the arrangement of electrons in the atom and in particular by the stability of the valence electron shell
- molecular formulae represent the number and type of atoms present in the molecules
- the type of bonding within ionic, metallic and covalent substances explains their physical properties, including melting and boiling points, conductivity of both electricity and heat and hardness
- the properties of covalent molecular substances, including low melting point, can be explained by their structure and the weak intermolecular forces between molecules; their non-conductivity in the solid and liquid/molten states can be explained by the absence of mobile charged particles in their molecular structure

6.1 Properties of non-metallic substances

In Chapter 4, you saw that the bonding between atoms in metallic elements is called metallic bonding. When metal atoms combine with atoms of non-metallic elements, the compounds contain another form of bonding, ionic bonding as described in Chapter 5. In this chapter, you will look at the chemical bonding that occurs when atoms of non-metals combine with each other.

EXAMPLES OF NON-METAL COMPOUNDS

Although there are fewer non-metals than metals in the periodic table, the atoms of non-metals form a much larger number of compounds than metals. Water, carbon dioxide, caffeine, sugar and cooking oil are just a few examples of common compounds formed from non-metals.

PROPERTIES OF NON-METALLIC ELEMENTS AND COMPOUNDS

The properties of compounds can tell you a lot about their chemical structures. Table 6.1.1 lists properties of a range of common substances. The substances chosen include a metal, ionic compounds and non-metal compounds. The information in Table 6.1.1 is useful because it allows you to:

- identify the characteristic properties of each category of substances; for example, metals conduct electricity as solids and liquids
- deduce information about the chemical structures of these substances. For example, if a substance has a high melting point, the bonding between particles must be strong. If a substance conducts electricity, it must contain charged particles that are free to move.

TABLE 6.1.1 Properties of some common metals, ionic compounds and non-metal compounds

Substance	Formula	Melting point (°C)	Boiling point (°C)	Conducts electricity as a solid?	Conducts electricity as a liquid?
copper	Cu	1084	2562	yes	yes
copper(II) chloride	CuCl ₂	498	993	no	yes
table salt (sodium chloride)	NaCl	801	1413	no	yes
laughing gas (nitrous oxide)	N ₂ O	-90	-88	no	no
water	H ₂ O	0	100	no	no
ammonia	NH ₃	-77	-33	no	no

From Table 6.1.1, you can see that:

- metals such as copper have high melting points and conduct electricity both as solids and as liquids. As you saw in Chapter 4, this indicates that strong bonds exist in metals and that they contain charged particles (delocalised electrons) that are free to move
- the ionic compounds copper(II) chloride and sodium chloride have high melting points and conduct electricity as liquids but not as solids. In Chapter 5, you learnt that ionic solids have strong bonds and that they contain charged particles (ions) that are free to move in a liquid form but not in a solid form

- the non-metal compounds ammonia, nitrous oxide (laughing gas) and water have low melting points and do not conduct electricity. Non-metal compounds can be soft in the solid state. Two conclusions about bonding in non-metal compounds that can be drawn from these properties are shown in Figure 6.1.1.

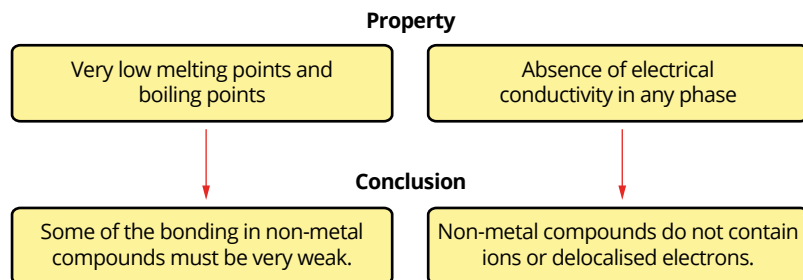
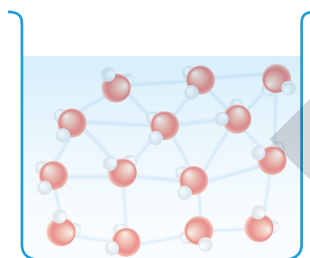


FIGURE 6.1.1 These conclusions can be drawn from the properties of non-metallic substances.

MOLECULES

You gain further insight into the chemical structure of non-metallic substances by considering what happens when you boil water in a beaker. Figure 6.1.2 shows that each water molecule contains two hydrogen atoms bonded to one oxygen atom. In between the molecule are weak forces holding the particles to each other.



Water in a beaker contains two types of attractive forces:

1. the bonds between hydrogen and oxygen atoms within each water molecule
2. the weak forces of attraction between the water molecules.

FIGURE 6.1.2 This representation shows the forces of attraction between molecules of water in a beaker.

When water boils (Figure 6.1.3), the weak forces of attraction between molecules must be overcome. However, the water molecules do not separate into hydrogen and oxygen. Rather, the water vapour that is formed still contains molecules in which two hydrogen atoms are bonded to one oxygen atom.

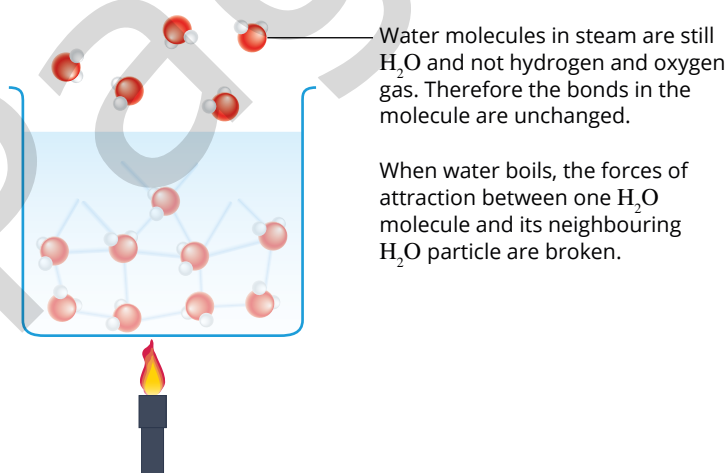


FIGURE 6.1.3 Changes occur to the forces of attraction between molecules in water when water starts to boil.

CHEMFILE

Big molecules with a spark

The covalent molecules we consider in this chapter are mostly small ones made of relatively few atoms such as water (H_2O), carbon dioxide (CO_2) and ammonia (NH_3), but molecules can have hundreds or even thousands of atoms. Plastics (polymers) such as polyvinyl chloride (PVC) are covalent molecular compounds with long chains of carbon atoms bonded together. These large molecular compounds generally have the properties we expect of molecular substances such as low melting points and non-conductors of heat and electricity but a new class of plastics is exciting much research interest. This class of plastics has a property we usually associate with metals—electrical conductivity.

Their electrical conductivity arises due to the particular way the atoms are bonded to each other. A sequence of alternating single and double bond bonds are formed (double bonds are considered in Section 6.2). This type of bonding creates a situation where electrons can move along the chain of carbon atoms. In effect, these electrons can be described as delocalised. These delocalised electrons can travel through the plastic and so allow the material to conduct electricity. There are many potential applications for these conducting plastics including solar cells, biosensors and printing electronic circuits.

i The valence shell is the highest energy shell (outer shell) of an atom that contains electrons.

The particles of water shown in Figure 6.1.3 are examples of a **molecule**. A molecule is a discrete (separate from others) group of atoms bonded together with a known formula.

Molecules can be found in some elements as well as in some compounds. The element oxygen, for example, forms O_2 molecules. Water can be referred to as a molecular compound with a **molecular formula** of H_2O . The forces of attraction between the hydrogen and oxygen atoms within water molecules are referred to as **intramolecular bonds**, or more simply bonds, and the forces of attraction between neighbouring water molecules are referred to as **intermolecular forces**.

The intramolecular bonds between the hydrogen and oxygen atoms in water molecules are strong compared to the intermolecular forces between the water molecules. It is the intermolecular forces that are broken when molecular substances such as water boil or melt. This allows the molecules to separate from each other while the atoms remain bound to one another within the molecules.

CHEMISTRY IN ACTION

Helium

The Sun is made up of about 24% helium and 75% hydrogen (Figure 6.1.4).

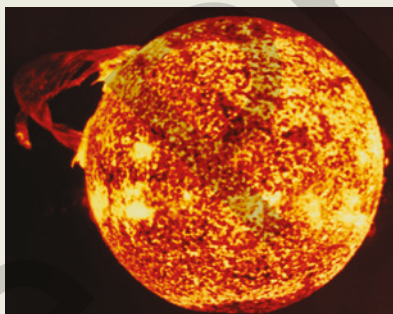


FIGURE 6.1.4 The Sun is made up of about 24% helium and 75% hydrogen. The first evidence for the existence of helium came from a spectral line observed in sunlight during a solar eclipse in 1868.

Helium is an interesting case study of bonding because there are almost no intermolecular forces between its atoms. Helium atoms are small and they have two electrons in their valence shell. The first shell can only hold two electrons, so this means the valence shell is full and the atoms are very stable. As a consequence, helium atoms do not form molecules.

The forces of attraction between helium atoms are almost non-existent. Helium has to be cooled to -269°C before it turns to a liquid. As a liquid, it can climb up the sides of its container and escape! Extraction

of helium takes advantage of its extremely low boiling point. Most helium is extracted by fractional distillation from natural gas which can have as much as 7% helium, as opposed to the approximate 5.2 ppm (parts per million by volume) in the atmosphere. As natural gas is cooled the other components liquefy and can be removed, leaving helium gas.

Although helium is the second most abundant element in the universe (second to hydrogen), it is very rare on Earth. It is used in liquid form (that is, at very low temperature) in Magnetic Resonance Imaging machines (MRIs) for medical purposes and particle accelerators such as the Large Hadron Collider, as well as in party balloons. Concerns about dwindling supplies of helium sparked debates about restricting its use to medical and scientific purposes—so no more party balloons. Present rates of use suggest supplies will last to about 2040, however, in 2016 a large reserve of helium was discovered in the Great Rift Valley in Tanzania which may have enough to extend reserves for one or two decades. As well, the methodology used to discover this reservoir has the potential to uncover other reserves in regions geologically similar to the Rift Valley.

6.1 Review

SUMMARY

- Many substances contain only non-metal atoms. Some of these substances are elements, while others are compounds.
- Non-metallic elements and compounds usually have low boiling points and do not conduct electricity.
- Many non-metallic elements and compounds are composed of molecules. Molecules are discrete groups of atoms of known formula, bonded together.
- There are two important types of forces of attraction in molecular compounds—the bonds within the molecules (intramolecular bonds) and intermolecular forces—the attractions between molecules.
- The melting and boiling points of non-metallic elements and compounds depend on the intermolecular forces between molecules. Generally, these molecular substances have low melting and boiling points and, when solid, the solids tend to be soft. This indicates they have weak forces of attraction between molecules.
- In general, non-metallic elements and compounds do not conduct electricity because they do not contain free-moving charged particles (neither delocalised electrons nor ions).

KEY QUESTIONS

- 1 Identify the non-metallic elements and compounds from this list: MgSO_4 , NO_2 , H_2 , Ni , CaCl_2 , Br_2 .
- 2 Write the definition of a molecule.
- 3 Explain the following general properties of non-metals.
 - a Non-metals do not conduct electricity.
 - b Non-metals have low melting and boiling points.
- 4 Identify whether each of the following statements about methane (CH_4) is true or false.
 - a Methane is a molecular compound.
 - b A molecule of methane contains four atoms.
 - c There are 20 atoms in four molecules of methane.
 - d The forces of attraction between the carbon and hydrogen atoms in methane are intermolecular forces.
- 5 When sugar is gently heated, it turns into a clear liquid. If the liquid is heated strongly, it turns black and a gas is produced. Explain what is happening to the two types of forces of attraction in sugar when it is heated. Use the terms 'intermolecular forces' and 'bonds' in your answer.

6.2 Covalent bonding

This section examines a series of simple molecules to help you to understand the concept of a **covalent bond**, which is formed when non-metallic atoms share electrons. Using your knowledge of the valence shell electron arrangements of non-metallic atoms, you will be able to predict the molecules that different elements can form.

COVALENT BONDS

When atoms bond, they become more stable if they gain a valence (outer) shell of eight electrons by combining with other atoms (the octet rule).

Commonly, when atoms of non-metals combine, electrons are shared so that each atom has eight electrons in its valence shell. Molecules formed in this way are more stable than the separate atoms.

Non-metallic atoms have a relatively high number of electrons in their valence shells and they tend to share rather than to transfer electrons. Covalent bonding occurs when electrons are shared between atoms. The positively charged nuclei of the neighbouring atoms each have an attraction for the shared electrons so this keeps the atoms held together in the molecule.

Single covalent bonds

When atoms share two electrons, one from each atom, the covalent bond formed is called a **single covalent bond**. Two examples of substances that contain single bonds are hydrogen and chlorine.

Example 1: Hydrogen

Hydrogen atoms have one electron. The valence shell for a hydrogen atom can hold a maximum of two electrons. A hydrogen atom can bond to another hydrogen atom to form a molecule of H_2 , as shown in Figure 6.2.1. It should be noted that because the valence shell for a hydrogen atom can hold a maximum of two electrons it does not follow the octet rule when forming bonds. It only needs two electrons to fill the valence shell.

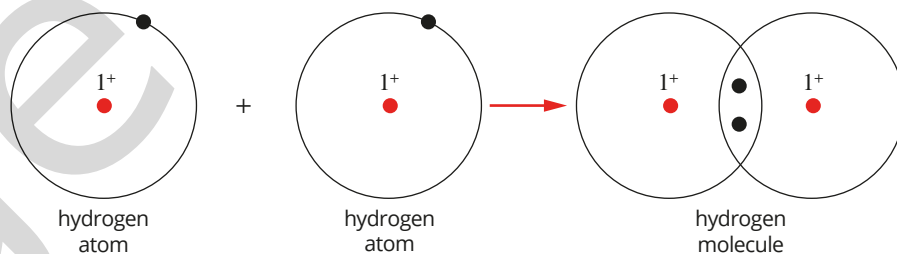


FIGURE 6.2.1 A covalent bond is formed when two hydrogen atoms share two electrons, one from each atom.

In the molecule that is formed:

- two hydrogen atoms share two electrons, one from each atom, to form a single covalent bond
- the atoms of hydrogen are now strongly bonded together by two electrons (an electron pair) in their valence shells.

The hydrogen molecule can be represented as H_2 . Molecules that contain two atoms like this are called **diatomic molecules**.

Two alternative ways of representing a hydrogen molecule are shown in Figure 6.2.2.

Remember that the nucleus of a hydrogen atom only contains a single proton. In a hydrogen molecule, the shared pair of electrons is attracted to nuclei of both hydrogen atoms. This means the two electrons will spend most of their time between the two nuclei instead of orbiting their own proton within its nuclei. This electrostatic attraction to the electrons keeps the molecule held together.

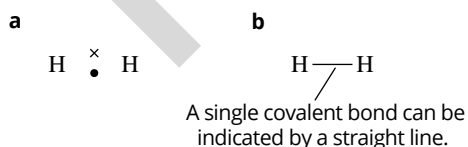


FIGURE 6.2.2 A hydrogen molecule can be represented by (a) showing the shared electrons as dots or crosses, or (b) showing the shared electron pair (the bond) as a line.

This model of a hydrogen molecule is consistent with the observed properties of hydrogen gas. The covalent bonds in the molecules are strong, but the intermolecular forces in hydrogen (the attractions between one molecule and the surrounding molecules) are weak. This explains the low melting temperature of -259°C for hydrogen. Hydrogen does not conduct electricity as it does not contain ions or delocalised electrons.

Hydrogen is an example of a covalent molecular substance.

Example 2: Chlorine

A chlorine atom has an electron configuration of 2,8,7. It requires one more electron to achieve eight electrons in its valence shell.

One chlorine atom can share an electron with another chlorine atom to form a molecule of chlorine with a single covalent bond. As a result, both atoms gain valence shells of eight electrons as shown in Figure 6.2.4. This is an example of the application of the octet rule.

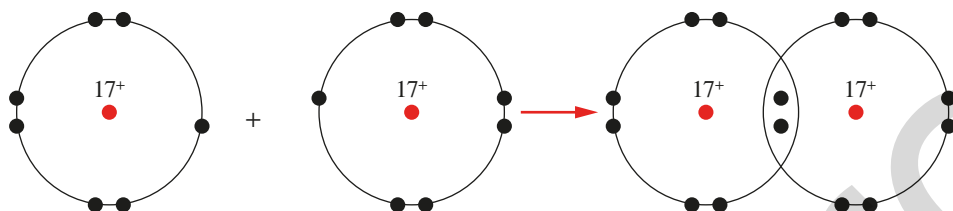


FIGURE 6.2.4 Two chlorine atoms share one electron each to form a chlorine molecule. Note that only valence-shell electrons are usually shown in these diagrams.

Lewis structure diagrams

Chemists often use **Lewis structure diagrams** (also known as **electron dot diagrams**). Lewis structure diagrams only show the valence shell electrons of an atom because only electrons in the valence shells of atoms are involved in bonding. In a Lewis structure diagram, lines can be used to represent the two electrons in a covalent bond. One line represents one pair of electrons.

The Lewis structure diagram also allows you to distinguish between bonding electrons and non-bonding electrons. A chlorine molecule has one pair of bonding electrons. The valence-shell electrons that are not involved in bonding are called the **non-bonding electrons**. Each chlorine atom has six non-bonding electrons, present as three pairs of electrons. Pairs of non-bonding electrons are known as **lone electron pairs** (or **lone pairs**).

Figure 6.2.5 shows two versions of a Lewis structure for a molecule of chlorine. It is easy to see that each chlorine atom in the molecule has eight electrons in its valence shell. In dot diagrams, you can use crosses in place of the dots (as in the case of diagram **a**) if you want to show which atoms have ‘provided’ the electrons.

Double covalent bonds

In a **double covalent bond**, two pairs of electrons (four electrons in total) are shared between the atoms, rather than just one pair.

For example, the oxygen molecule contains a double covalent bond. The electron configuration of an oxygen atom is 2,6. Each oxygen atom requires two electrons to gain a stable valence shell containing eight electrons. Therefore, when one oxygen atom bonds to another one, each atom shares two electrons.

CHEMFILE

Hydrogen airships

Hydrogen has a low density. This was once thought to make it suitable for use in airships. Zeppelins were a type of rigid airship that was used as a mode of transport during the early 1900s. However, their popularity as a way of travel decreased after the hydrogen gas in the zeppelin *Hindenburg* (Figure 6.2.3) caught fire in 1937, killing many on board.



FIGURE 6.2.3 The German passenger zeppelin *Hindenburg* exploded during its attempt to dock at the Lakehurst Naval Air Station in the United States.

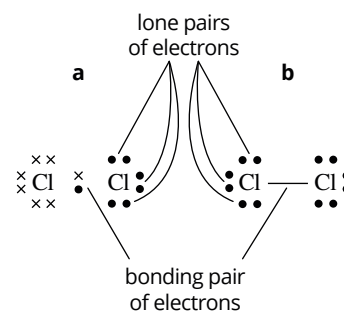


FIGURE 6.2.5 Two different ways of representing the chlorine molecule. They show the valence-shell electrons only. (a) All electrons are shown by dots or crosses. (b) A line is used to represent the pair of electrons that forms the covalent bond. The dots are the non-bonding electrons.

CHEMFILE

The strong triple covalent bond in N₂

The triple covalent bond in nitrogen gas (N₂) is relatively strong and not easily broken. This means that nitrogen gas is relatively unreactive. Nitrogen is an essential element in living organisms because it is a major component of proteins and other biological molecules. Even though 78% of air is nitrogen gas, very few organisms can make use of the nitrogen because it is so unreactive. Only nitrogen-fixing microorganisms are able to convert nitrogen gas into soluble nitrogen-containing compounds. These compounds are then absorbed by plants, allowing nitrogen to be passed up the food chain.

In the early 20th century, German chemist Fritz Haber invented a process for converting nitrogen gas and hydrogen gas into ammonia, which is used to make synthetic fertilisers. This enabled more food to be grown to feed a growing world population.

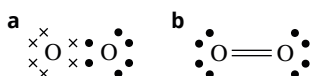


FIGURE 6.2.7 (a) The Lewis structure diagram shows that O₂ has a double covalent bond. Four electrons are shared and each oxygen has two non-bonding electron pairs. (b) Alternatively, the bonding electrons are represented by two lines, one line per bond (pair).

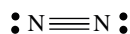


FIGURE 6.2.9 The triple covalent bond in a nitrogen molecule can be shown as three parallel lines.

As you can see in Figure 6.2.6, each oxygen atom in the molecule now has eight valence-shell electrons, four of these are bonding electrons and four are non-bonding electrons.

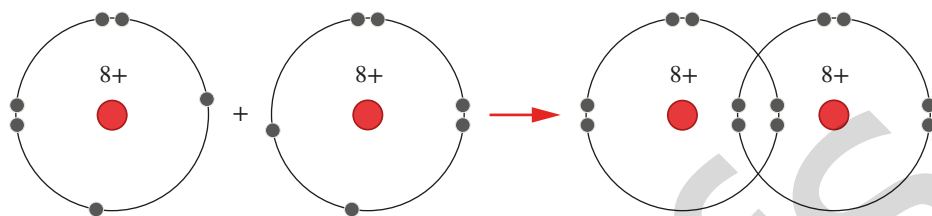


FIGURE 6.2.6 In oxygen molecules, each oxygen atom contributes two electrons to the bond between the atoms.

In Figure 6.2.7, you can see two different versions of Lewis structures that represent the oxygen molecule.

Triple covalent bonds

A **triple covalent bond** occurs when three electron pairs are shared between two atoms. For example, the nitrogen molecule contains a triple bond. The electron configuration of nitrogen is 2,5. A nitrogen atom requires three electrons to achieve eight electrons in its valence shell. When it bonds to another nitrogen atom, each atom contributes three electrons to the bond that forms, as shown in Figure 6.2.8. The Lewis structure of the nitrogen molecule is shown in Figure 6.2.9.

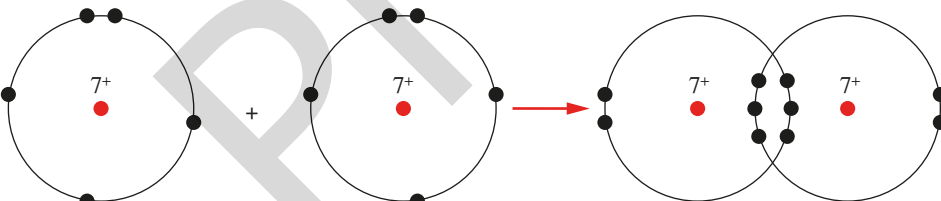


FIGURE 6.2.8 Nitrogen atoms contribute three electrons each to form a triple covalent bond in a molecule of N₂.

MOLECULAR COMPOUNDS

A diatomic molecule contains two atoms. The molecules discussed so far have been diatomic molecules that contain atoms of the same elements.

Covalent bonds can also form between atoms of different elements. Hydrogen chloride (HCl) is a simple example (Figure 6.2.10). A hydrogen atom requires one electron to gain a stable valence shell, as does a chlorine atom. They can share an electron each and form a single covalent bond.

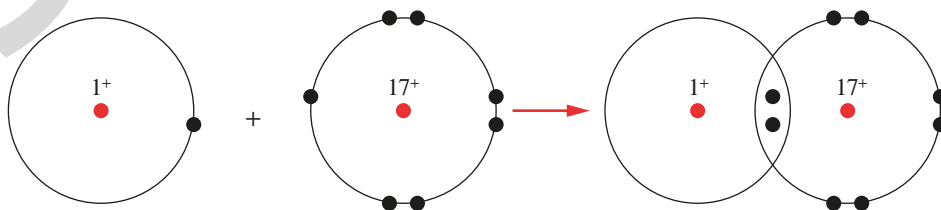


FIGURE 6.2.10 A pair of electrons is shared in the formation of a molecule of HCl.

Polyatomic molecules

If the atoms of one element have a different number of valence electrons from the atoms of the element it is bonding with, the molecule that is formed may not be a simple diatomic molecule. Molecules made up of more than two atoms are called **polyatomic molecules**. Two examples of polyatomic molecules are water and methane.

Example 1: Water

When a compound forms between hydrogen and oxygen, an oxygen atom shares two electrons and a hydrogen atom shares one. To resolve this imbalance, two hydrogen atoms each share one electron with an oxygen atom.

As you can see in the Lewis structure shown in Figure 6.2.11, a water molecule contains:

- two single covalent bonds, each containing a shared electron pair
- four non-bonding electrons on the oxygen atom.

Example 2: Methane

When a compound forms between carbon and hydrogen, four hydrogen atoms are needed to provide the four electrons required in order to have eight electrons in the valence shell of a carbon atom (Figure 6.2.12). The molecule formed has a chemical formula of CH_4 and is called methane.

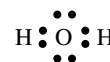


FIGURE 6.2.11 A water molecule has two single covalent bonds. Electrons can be represented as dots and crosses, or simply, as here, by dots.

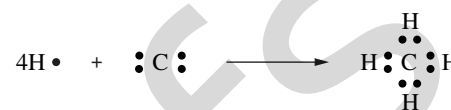


FIGURE 6.2.12 In a methane molecule, a carbon atom shares one electron with each of four hydrogen atoms to gain eight electrons in its valence shell.

EXTENSION

Polyatomic ions

We have considered Lewis structures for polyatomic molecules but there are also polyatomic ions, for example hydroxide ion (OH^-), ammonium ion (NH_4^+), nitrate ion (NO_3^-) and sulfate ion (SO_4^{2-}). Polyatomic ions are made up of two or more atoms. The atoms in a polyatomic ion bond to each other covalently in the same way as atoms in a polyatomic molecule and can be represented by Lewis structures.

When drawing the Lewis structure for a polyatomic ion, the number of electrons available for sharing is determined from the number of valence electrons from each atom in the ion as well as the electrons either gained or lost as indicated by the charge on the ion. For example, in the hydroxide ion the oxygen atom has six valence electrons and the hydrogen has one. Hydroxide has a charge of negative one so it has gained an electron. This means there are a total of eight electrons to share between the oxygen and hydrogen atoms. The hydrogen atom needs one more electron to complete its valence shell (remember it only needs two electrons) and the oxygen atom needs two electrons to complete its valence shell (oxygen follows the octet rule). Thus, the oxygen atom and hydrogen atom share a pair of electrons to form a single bond. Its Lewis structure is shown in Figure 6.2.13. Note the eight electrons in the valence shells of the atoms.



FIGURE 6.2.13 Lewis structure for the hydroxide ion. A single covalent bond forms between the atoms.

Consider the ammonium ion. Nitrogen has five valence electrons and each hydrogen has one valence electron (thus four electrons from the hydrogen atoms). The +1 charge tells us that when the atoms bonded to form the ion one electron was lost, so there are eight valence electrons available for bonding. Figure 6.2.14 shows the central nitrogen atom sharing a pair of electrons with each of the four hydrogen atoms.

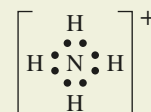


FIGURE 6.2.14 Lewis structure for the ammonium ion. The central N atom has an octet and each H atom has a pair of electrons.

Similarly, NO_3^- has 24 valence electrons available for sharing and oxygen atoms form a double bond and two single bonds around the central N atom (Figure 6.2.15).



FIGURE 6.2.15 Lewis structure for the nitrate ion. The central N atom forms a single bond with two oxygen atoms and a double bond with one oxygen atom. All atoms have an octet.



FIGURE 6.2.16 H_2 and HCl both contain only one bond, so they are linear in shape.

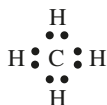


FIGURE 6.2.17 The carbon atom in methane has four pairs of electrons in its valence shell. These electron pairs repel each other.

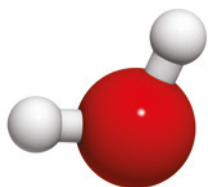


FIGURE 6.2.19 Ball-and-stick model of a water molecule



FIGURE 6.2.20 Space-filling model of a water molecule

SHAPES OF MOLECULES

The shapes of diatomic molecules such as H_2 and HCl are simple. With two atoms only, a diatomic molecule is linear, as you can see in Figure 6.2.16.

Lewis structures can be used as a starting point in determining the shape of molecules. These diagrams show the valence-shell electron pairs in the molecule. Electron pairs have a negative charge and they repel each other. The electron pairs arrange themselves as far away from each other as possible.

In the case of methane, the Lewis structure diagram (Figure 6.2.17) shows four valence-shell electron pairs around the carbon atom. Repulsion between the electron pairs means that the hydrogen atoms form a **tetrahedral shape** around the carbon atom (Figure 6.2.18). In this molecule, the bonds between the carbon and hydrogen atoms are at an angle of 109.5° to each other.

You will have a more detailed look at the shapes of molecules in Chapter 12.

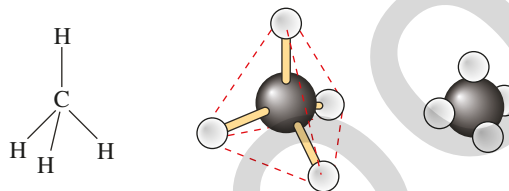


FIGURE 6.2.18 Three types of 3-dimensional representations of a methane molecule show that hydrogen atoms form a tetrahedral arrangement around the central carbon atom.

Representations of molecules

A Lewis structure diagram does not show the shape of a molecule but there are several other accepted styles for representing molecules that do. A **ball-and-stick model**, as shown in Figure 6.2.19, displays the shape and the type of bonds (single, double or triple) in a molecule. The atoms are represented by spheres that are connected by rods, representing the bonds. The spheres represent the centre of the atom, not the whole space the atom occupies.

A **space-filling model** (Figure 6.2.20) also uses spheres but this time the spheres represent the whole atom, including its electron charge cloud. Different coloured spheres are used in both types of models to identify the different elements.

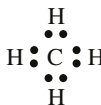
Each different representation of a molecule has its advantages. A Lewis structure diagram is often drawn to determine the number and type of bonds present, while the ball-and-stick and space filling models show the shape of the molecule and the position of atoms. Table 6.2.1 compares different representations of a PF_3 molecule.

TABLE 6.2.1 Comparison of different representations of phosphorus trifluoride, PF_3

	Lewis structure with electrons as dots	Lewis structure with lines for bonds	Ball-and-stick	Space-filling
Diagram				
Model best used for	determining the formula of the molecule and the type of bonds present	simplifying Lewis structure diagram	displaying the molecule shape	showing the relative size and position of the atoms in the molecule
Limitation of model	does not show the relative size of atoms or shape of the molecule	does not show the relative size of atoms or shape of the molecule	shows the shape but not the relative sizes of the atoms	shows the relative size and position of the atoms but does not show bond angles or types of bonds

Worked example 6.2.1

LEWIS STRUCTURE DIAGRAMS

Draw a Lewis structure diagram of methane (CH_4).	
Thinking	Working
Write the electron configuration of the atoms in the molecule.	C electron configuration: 2,4 H electron configuration: 1
Determine how many electrons each atom requires for a stable valence shell.	C requires 4 electrons. H requires 1 electron.
Draw a Lewis structure diagram of the likely molecule, ensuring that each atom has a stable valence shell. Electrons not involved in bonding will be in non-bonding (lone) pairs.	Draw a Lewis structure diagram of the molecule. 

Worked example: Try yourself 6.2.1

LEWIS STRUCTURE DIAGRAMS

Draw a Lewis structure diagram of ammonia (NH_3).

NON-OCTET MOLECULES

As you have seen, when hydrogen atoms form covalent bonds, they only need two electrons to complete their valence shell. The atoms beryllium (Be) and boron (B) only have two or three electrons respectively in their valence shells. Therefore, when these electrons 'pair-up' to form bonds, beryllium is limited to a total of four electrons and boron six electrons in the outer shell. However, due mainly to the small size of these atoms, this arrangement is stable enough for the molecules to form, even though they do not follow the octet rule.

For example, for boron trifluoride (BF_3), each fluorine atom (electron configuration 2,7) forms one covalent bond. Boron has an electron configuration of 2,3. There are only three electrons in the valence shell of boron. Each of these is shared with one of three fluorine atoms to form covalent bonds. This results in only three pairs of electrons (six electrons) around the boron atom in BF_3 (Figure 6.2.21). Boron is said to have an incomplete octet but the molecule still forms.

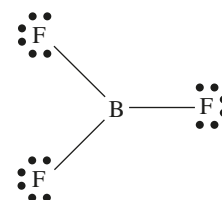


FIGURE 6.2.21 Lewis structure of BF_3

6.2 Review

SUMMARY

- The atoms in molecules are held together by covalent bonds.
- A covalent bond involves the sharing of electrons.
- Covalent bonds form between non-metallic atoms, often enabling the atoms to gain valence shells containing eight electrons (except hydrogen which gains a valence shell containing two electrons).
- A single covalent bond forms when two atoms share a pair of electrons.
- A double covalent bond forms when two atoms share two pairs of electrons.
- A triple covalent bond forms when two atoms share three pairs of electrons.
- Valence-shell electrons that are not involved in bonding are called lone pairs.
- Lewis structure (electron dot) diagrams show the valence electron arrangements of atoms in a molecule.

KEY QUESTIONS

- How many covalent bonds are formed between atoms in these diatomic molecules?
 - H_2
 - N_2
 - O_2
 - F_2
- Draw Lewis structure diagrams for each of the following molecules.
 - fluorine (F_2)
 - hydrogen fluoride (HF)
 - water (H_2O)
 - tetrachloromethane (CCl_4)
 - phosphine (PH_3)
 - butane (C_4H_{10})
 - carbon dioxide (CO_2)

You can use lines to show bonding pairs of electrons if you wish.
- What is the maximum number of covalent bonds an atom of each of the following elements can form?
 - F
 - O
 - N
 - C
 - H
 - Ne
- When oxygen forms covalent molecular compounds with other non-metals, the Lewis structures that represent the molecules of these compounds all show each oxygen atom with two lone pairs of electrons. Why are there always two lone pairs?
- Suggest the most likely formula of the compound formed between the following pairs of elements.
 - C, Cl
 - N, Br
 - Si, O
 - H, F
 - P, F
- Identify the type of representation of a molecule that best fits the following descriptions.
 - shows the shape of the molecule and type of atoms present
 - shows the arrangement of valence electrons between atoms in a molecule as dots
 - shows the relative size and position of atoms and the shape of molecule

Chapter review

KEY TERMS

ball-and-stick model
covalent bond
diatomic molecule
double covalent bond
electron dot diagram
intermolecular force
intramolecular bond

Lewis structure diagram
lone electron pair
molecular formula
molecule
non-bonding electron
polyatomic molecule
single covalent bond

space-filling model
tetrahedral shape
triple covalent bond

Properties of non-metallic substances

- 1 Hydrogen chloride (HCl) exists as a gas at room temperature. What can you conclude about the strength of the intermolecular forces in pure hydrogen chloride?
- 2 Non-metallic substances generally do not contain free-moving charged particles. What does this indicate about the general physical properties of non-metallic substances?
- 3 Solid carbon dioxide sublimates when it is heated; that is, it goes straight from a solid to a gas. Explain what happens to the forces of attraction in carbon dioxide when it sublimates. Use the terms 'intermolecular forces' and 'bonds' in your answer.
- 4 Explain why most substances that have an odour are covalent molecular substances.
- 5 Use the properties of the four substances given in the table below to determine which of A, B, C and D are magnesium chloride, copper, naphthalene or phosphorus trihydride. Give a brief reason for your decision.

Substance	Electrical conductivity in aqueous solution	Melting point (°C)	Hardness when solid
A	no	80	soft
B	yes	714	brittle
C	not soluble in water	1084	hard
D	no	-133	soft

Covalent bonding

- 6 Select the statement that best describes the way hydrogen atoms bond to each other.
A One hydrogen atom donates an electron to another hydrogen atom to form a molecule.
B Hydrogen atoms form a lattice with delocalised electrons.
C Hydrogen atoms share electrons to obtain a complete valence shell of eight electrons.
D Two hydrogen atoms share an electron each to form a hydrogen molecule.
- 7 In a molecule of nitrogen (N₂), how many of the following are there?
a bonding electrons
b covalent bonds
c non-bonding electrons
- 8 The formula of a molecule is XY₄. Select the alternative that could match this formula.
A OH₄
B CH₄
C HBr₄
D CO₄
- 9 Atom X has an electron configuration of 2,6. Atom Y has a configuration of 2,7. What is the likely molecular formula of the compound that will form between X and Y?
- 10 Silicon forms a compound with element X with formula SiX₄. Give the most likely number of valence electrons in atoms of X so that the bonding follows the octet rule. Explain your answer.
- 11 Explain why neon atoms do not form covalent bonds.

- 12** How many lone pairs are in each of the following molecules?
- H_2
 - NH_3
 - HCl
 - O_2
- 13** Draw Lewis structures (electron dot diagrams) for each of the following molecules and identify the number of bonding and non-bonding electrons in each molecule.
- HBr
 - H_2O
 - CF_4
 - C_2H_6
 - PF_3
 - Cl_2O
 - CH_4
 - H_2S
- 14** The atoms in molecules of nitrogen, oxygen and fluorine are held together by covalent bonds. How are the bonds in these molecules:
- similar?
 - different?
- 15** Experimental evidence shows that the double bond between the two oxygen atoms in O_2 is much stronger than a single bond between two oxygen atoms in a compound such as hydrogen peroxide (H_2O_2).
- Draw Lewis structure diagrams and structural formulae for O_2 and H_2O_2 .
 - Explain why the oxygen double bond is stronger than the oxygen single bond.
 - Why does oxygen not form a triple bond or three single covalent bonds?

Connecting the main ideas

- 16** Classify the following substances as metallic, ionic or molecular: CuCl_2 , Ag , HCl , H_2O , Cu , CaS , NH_3 .
- 17** The labels have fallen off the jars of two white powders in a laboratory. One jar contains the ionic compound calcium chloride and the other the covalent molecular compound glucose. Describe a simple laboratory test that can be done to identify the powders. Both are soluble in water.
- 18** Describe how the atoms in a covalent molecule are held together.
- 19** Identify whether each of the following statements about carbon dioxide (CO_2) is true or false.
- Carbon dioxide is a molecular compound.
 - A molecule of carbon dioxide contains three atoms.
 - There are 99 oxygen atoms in 33 molecules of carbon dioxide.
 - The forces of attraction between the carbon and oxygen atoms in carbon dioxide are covalent bonds.
 - There are two single covalent bonds in a molecule of carbon dioxide.
 - There are four lone pairs of electrons in a molecule of carbon dioxide.